

Task 1: A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

1. **Check the power adapter and outlet** – Test the outlet with another device and inspect the adapter's LED and cable for damage or a loose DC jack connection.
2. **Try a different power charger** – If a compatible spare is available, swap it in to see whether the original adapter has failed internally, which is a common cause.
3. **Remove the battery and try AC-only power** – A dead or swollen battery can prevent the system from powering on even on AC power; if it boots with the battery out, the battery is the cause.
4. **Do a hard reset / drain residual power** – Unplug everything, remove the battery if possible, hold the power button for 15-30 seconds to discharge capacitors, then reconnect and try powering on to clear minor power-circuit glitches.
5. **Check for any signs of life** – No response at all points toward a motherboard or charging circuit fault, while lights or fan spinning but no display suggests the issue lies elsewhere (display or RAM), not the power system.

Task 2: You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

1. **Confirm the system is actually running** – Listen for fan noise, check the power/HDD LED, and check if Caps Lock toggles as a live-system indicator.
2. **Check the monitor itself** – Make sure the monitor is powered on and set to the correct input source (HDMI/VGA/DisplayPort); try unplugging and replugging its power cable.
3. **Check the video cable connection** – Reseat the cable at both ends and try a different cable if available, since damaged cables are a very common cause.
4. **Test with a different monitor or port** – Try another port on the GPU or integrated graphics card or connect a spare monitor to isolate whether the fault is the monitor or the PC's video output.
5. **Check for external GPU vs onboard display** – Make sure the cable is plugged into the dedicated GPU port, not the motherboard's onboard port, which is a very common mistake.
6. **Listen for beep codes** – Many motherboards beep in patterns to indicate a RAM or GPU fault if the display never initializes.
7. **Reseat the RAM and GPU** – Power off, unplug, and reseat the memory sticks and graphics card, since a loose connection can cause no display despite the system powering on.

8. **Boot into Safe Mode or check via remote access** – If you can hear Windows startup sounds or are able to access the PC remotely, the OS is fine and the issue is purely display-related, pointing to drivers or hardware output rather than the OS.

Task 3: A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

1. **Check the mouse's power/battery** – Swap in fresh batteries or fully charge the mouse if rechargeable, since a dead battery is the most common cause.
2. **Check the receiver/dongle connection** – Unplug and replug the USB receiver, ideally into a different USB port, to rule out a faulty or loose port.
3. **Test the mouse on another computer** – If it doesn't work there either, the mouse itself is likely faulty; if it works fine, the issue is isolated to the PC.
4. **Test a different mouse on this same USB port** – If another mouse works fine, that confirms the port and system are okay, and the problem is isolated to the original mouse or its receiver.
5. **Check Device Manager for driver issues** – Look for a yellow warning icon next to the mouse or HID device entries, and update or reinstall the driver if there is a conflict.
6. **Check system Bluetooth/USB settings** – Confirm a Bluetooth mouse is actually paired with the correct device, and check power management settings, since Windows sometimes powers down USB ports to save energy.

Task 4: Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

1. **Dust buildup in fans and vents** – Open the case and visually inspect the CPU fan, GPU fan, and intake/exhaust vents for blockages, since heavy dust traps heat. Fix by cleaning with compressed air, holding fan blades still while spraying to avoid bearing damage.
2. **Dried-out or poorly applied thermal paste** – Check CPU temperatures at idle and under load using a monitoring tool (such as HWMonitor); very high idle temps or a big spike under load suggests poor heat transfer. Fix by removing the heatsink, cleaning off old paste, and reapplying a fresh, even layer.
3. **Faulty or insufficient fan/airflow** – Check whether all case fans and the CPU cooler fan are spinning and reaching expected RPM, using motherboard utilities or BIOS. Fix by replacing the faulty fan and ensuring the case has enough intake/exhaust fans for a gaming workload.